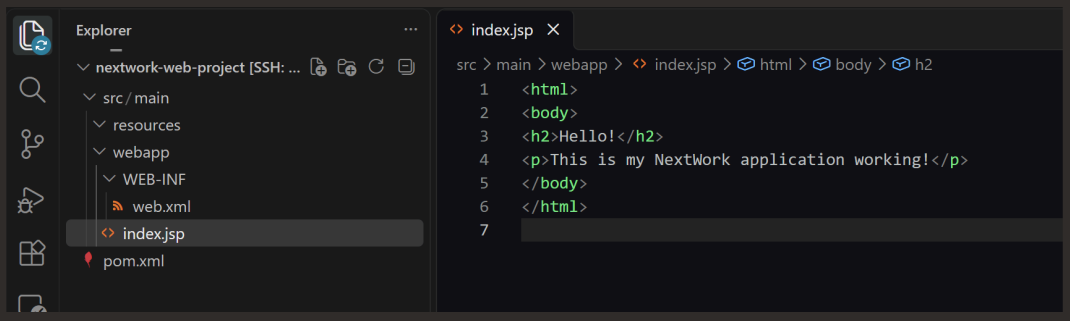
SET UP A WEB APP IN THE CLOUD

By Maryna Maksimuk | October 2026



Introducing Today's Project!

This project represents Day One of a 6-Day DevOps Challenge. The goal of this project is to build a complete CI/CD pipeline while developing the application from the ground up.

We will begin by launching an AWS EC2 instance and connecting to it through VS Code. From there, we will create and configure a web application, establish the required infrastructure, and prepare the project for automated build, testing, and deployment through a CI/CD pipeline.

I plan to complete the next project tomorrow as part of my 6-Day DevOps Challenge.

I completed this project to advance my DevOps career, strengthen my hands-on technical skills, and demonstrate my knowledge and practical experience with AWS, Linux, Java, Maven, SSH, and CI/CD concepts.

Key tools and concepts

The key services and concepts I learned in this project included AWS EC2 for cloud infrastructure, SSH and key pairs for secure server access, VS Code and Remote SSH for remote development, Java and Apache Maven for application development and builds, and the fundamentals of setting up a CI/CD pipeline for automated application delivery.

Project reflection

It took me approximately 2 hours to complete the project, including setting up the EC2 instance, configuring SSH and VS Code, creating the Java application with Maven, and exploring the application files.

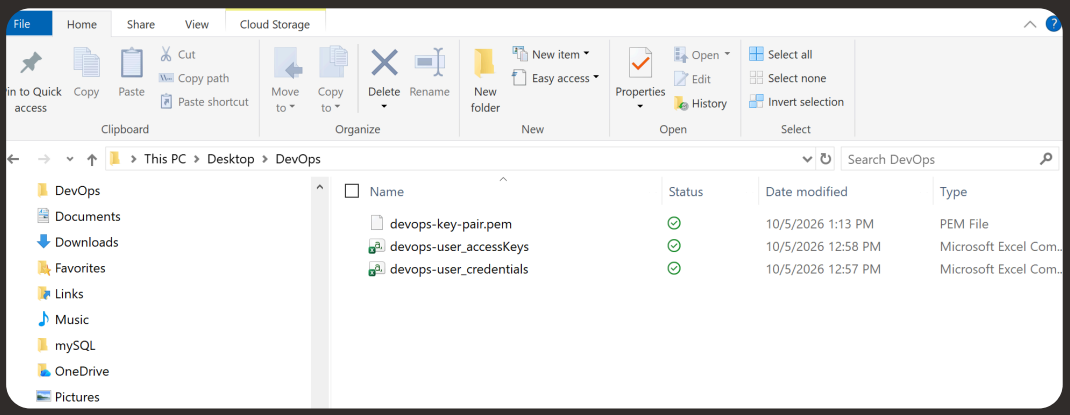
One thing I didn't expect was being able to edit code stored directly on the EC2 instance through VS Code using Remote - SSH, as if the files were located on my local computer.

Launching an EC2 instance

What I did in this step

In this step, I will launch and configure an AWS EC2 instance to serve as the cloud-based development environment, including setting up secure key-pair authentication and the necessary network settings.

I started this project by launching an AWS EC2 instance to provide a cloud-based environment where I can develop, configure, and deploy the web application and build the CI/CD pipeline.



I also enabled SSH

SSH is a secure protocol that allows me to remotely connect to and manage a server over an encrypted connection.

Key pairs

An AWS key pair provides secure SSH access to an EC2 instance, using a private key stored on my computer and a corresponding public key stored on the instance.

Downloaded key pair file

Once I set up my key pair, AWS automatically downloaded .pem file

Set up VS Code

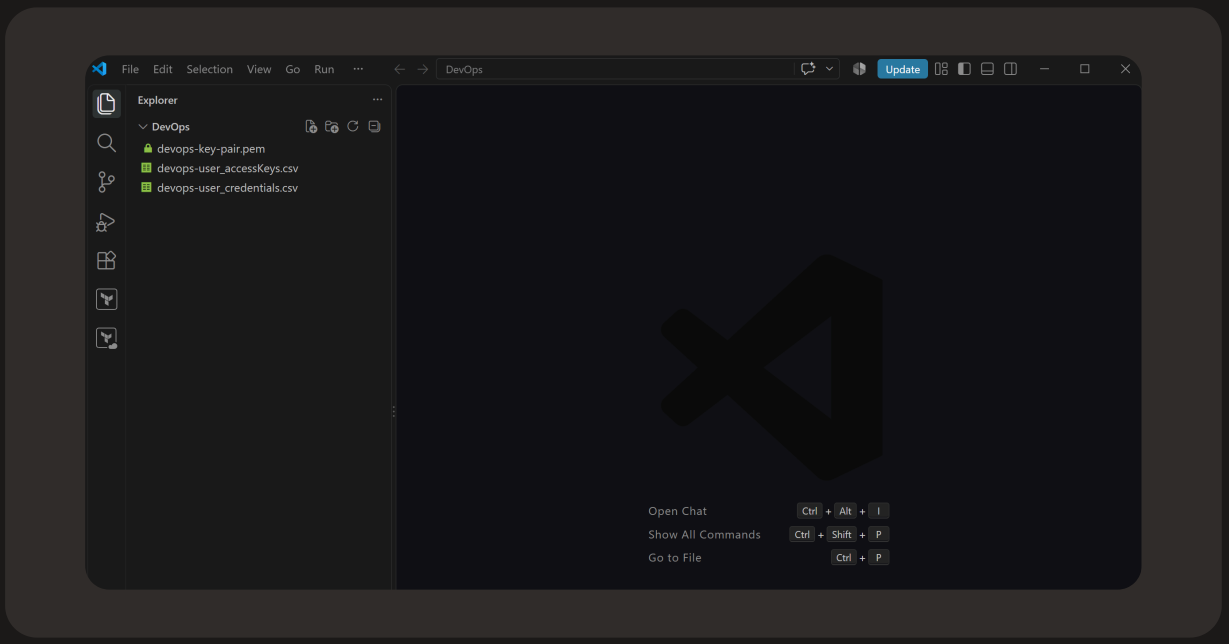
What I did in this step

In this step, I will install and configure VS Code to connect to the EC2 instance, set up its terminal, and configure the SSH key permissions for secure access.

What is VS Code?

VS Code is a free code editor that allows developers to write, edit, and manage application code and connect to remote servers such as AWS EC2 instances.

I will use VS Code to connect to my AWS EC2 instance, create and edit the web application code, and manage the project through its integrated terminal.

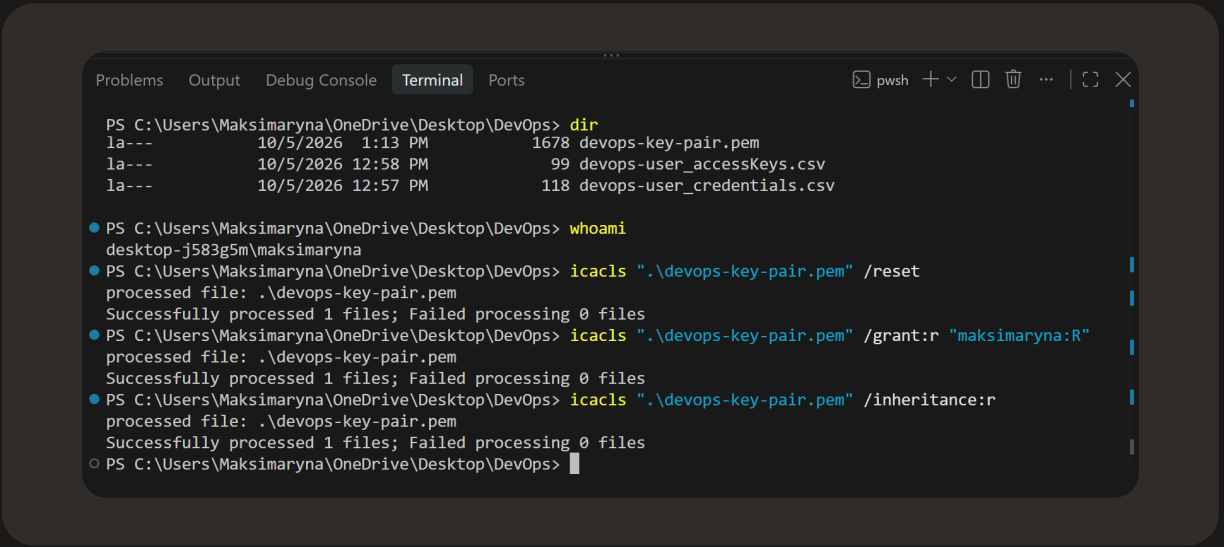


My first terminal commands

The VS Code terminal is a built-in command-line interface that allows me to run commands, manage files, and connect to and manage my AWS EC2 instance directly from VS Code. The first command I ran for this project was dir

Updating file permissions

I secured the .pem file by using the Windows icacls commands. I used /reset to remove the existing permissions, /grant:r "USERNAME":R to give only my user account read access, and /inheritance:r to disable inherited permissions from the parent folder. This ensured that the private key could only be accessed by my user account.



SSH connection to EC2 instance

What I did in this step

In this step, I will connect to my AWS EC2 instance securely using SSH, allowing me to access the remote server and begin setting up and developing the web application.

Connecting to EC2

I connected to the AWS EC2 instance using SSH with the instance's IPv4 DNS address and the private .pem key, using the command ssh -i "my-key.pem" ubuntu@<EC2-IPv4-DNS>.

This command required an IPv4 address

A server's IPv4 DNS is a human-readable hostname that points to the server's IPv4 address, making it easier to connect to the server without using its numeric IP address directly.



Maven & Java

What I did in this step

In this step, I will install Apache Maven and Amazon Corretto 8 (Java) on my EC2 instance and verify that both installations are working correctly.

Why I'm using Maven

Apache Maven is a build and project management tool for Java applications that helps automate tasks such as compiling code, managing dependencies, running tests, and packaging the application.

Maven is required in this project to build the Java web application, manage its dependencies, run tests, and package it so it can be deployed through my CI/CD pipeline.

Why I'm using Java

Java is a programming language and runtime platform used to build and run applications, and in this project, it provides the environment needed to build and run our Java web application.

I am using Java because the web application is Java-based, and I use Apache Maven to build the application, manage its dependencies, run tests, and package it for deployment through the CI/CD pipeline.

Create the Application

What I did in this step

In this step, I will use Maven commands in the terminal to generate the Java-based web application that will be used in the project.

Creating the Java web app

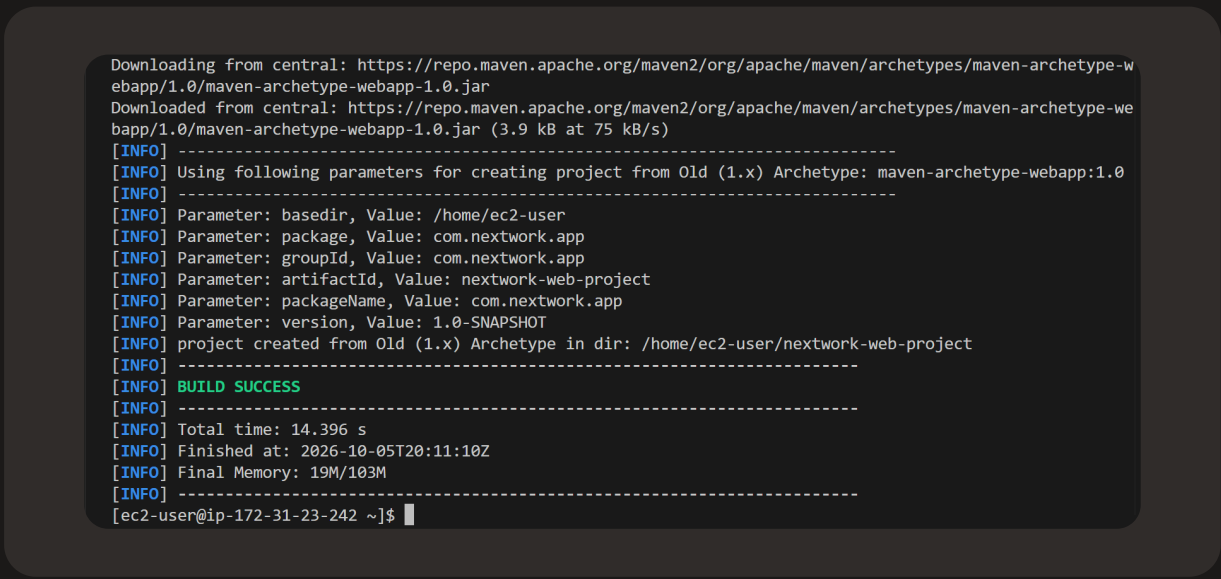
I created the Java web application using Apache Maven's mvn archetype:generate command, which generated the project structure, configuration files, and initial source code needed for the application.

Installing Remote - SSH

I installed Remote - SSH to securely connect VS Code to my EC2 instance and edit and manage the Java web application files directly on the remote server without having to copy them back and forth.

SSH configuration details

The SSH configuration file contained the connection details for my EC2 instance, including the host name, EC2 IPv4 DNS address, username, and the path to the private .pem key used for authentication.

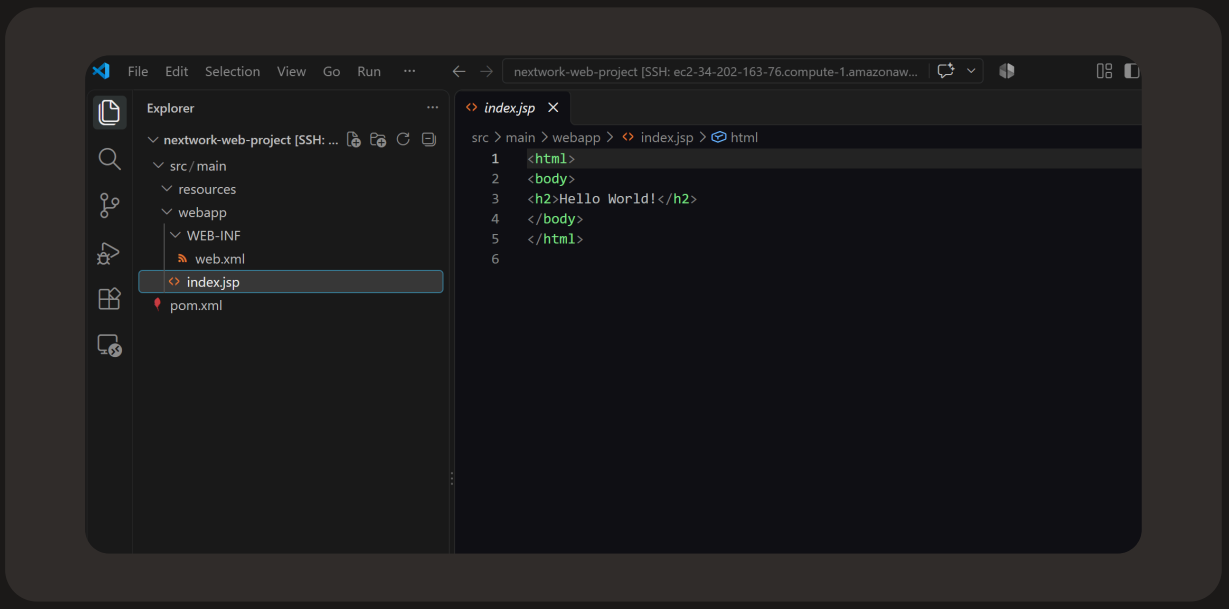


Create the Application

Exploring the project structure

In the VS Code File Explorer, I could see the Java web application's project structure and files stored on my EC2 instance, including the source code, pom.xml Maven configuration file, and project directories such as src/main and src/test.

1) src is the main source directory where the application's Java code and other project files are stored.
2) webapp is the folder that contains the web application's files, such as HTML, JSP pages, CSS, JavaScript, and other web resources that are served to users.



Using Remote - SSH

What I did in this step

In this step, I will install the Remote SSH extension in VS Code, connect VS Code to my EC2 instance, and use it to explore and edit the files of my Java web application directly on the server.

Updating the web app

index.jsp is the main JSP (JavaServer Pages) file that serves as the default web page of the Java application, containing the HTML and Java-based content that users see when they access the application.

I edited index.jsp directly in VS Code by opening the file through the File Explorer, modifying its HTML/JSP content, and saving the changes to the EC2 instance.

